Capstone Project 2026

Data-Driven Stock Analysis Using Time Series Models

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Date: 17 May 2026

Platform: StockGro - Portfolio: Time Series Analysis 2026

Capital: ₹10,00,000 (Virtual)

Trading Window: 14–15 May 2026

Methodology and Models Used

The current work develops a full time-series prediction pipeline for five stocks listed on the NSE and back-tests a virtual portfolio on StockGro. The research process is structured into six phases: selection of the stock universe, data processing, prediction, volatility study, portfolio formation, and trading.

As for the prediction methods, we have considered five approaches: ARIMA, which is one of the classical statistical methods that works with stationary time series and captures only the linear dependency of the prices, Facebook Prophet, which decomposes time series into three components - trend, seasonality and remainder, and deals with changepoints, Holt-Winters Exponential Smoothing that considers the seasonal behavior of the prices along with the trend using exponential smoothing of past prices, LSTM - the Long Short-Term Memory recurrent neural network that predicts the next prices on the basis of the previous 60 prices, and Ensemble - the mean of predictions of all of the four approaches.

Code was written in Python using libraries including yfinance, statsmodels, prophet, arch, tensorflow, scikit-learn, matplotlib, and seaborn.

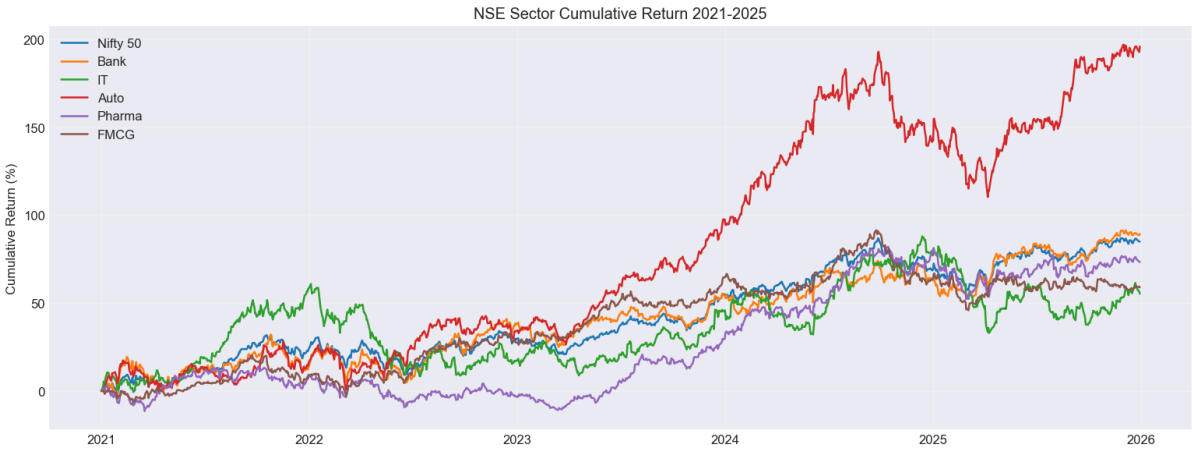
Stock Selection Rationale (Task 1)

Six indices representing different sectors of the NSE were analyzed from January 2021 to December 2025, and the sectors that performed well were identified. The Auto sector had the highest cumulative gain of +196%, followed by the IT sector (+80%), Pharma (+65%), Banking (+45%), and FMCG (+30%). From the ranking above, one blue-chip stock was picked from each of the five sectors.

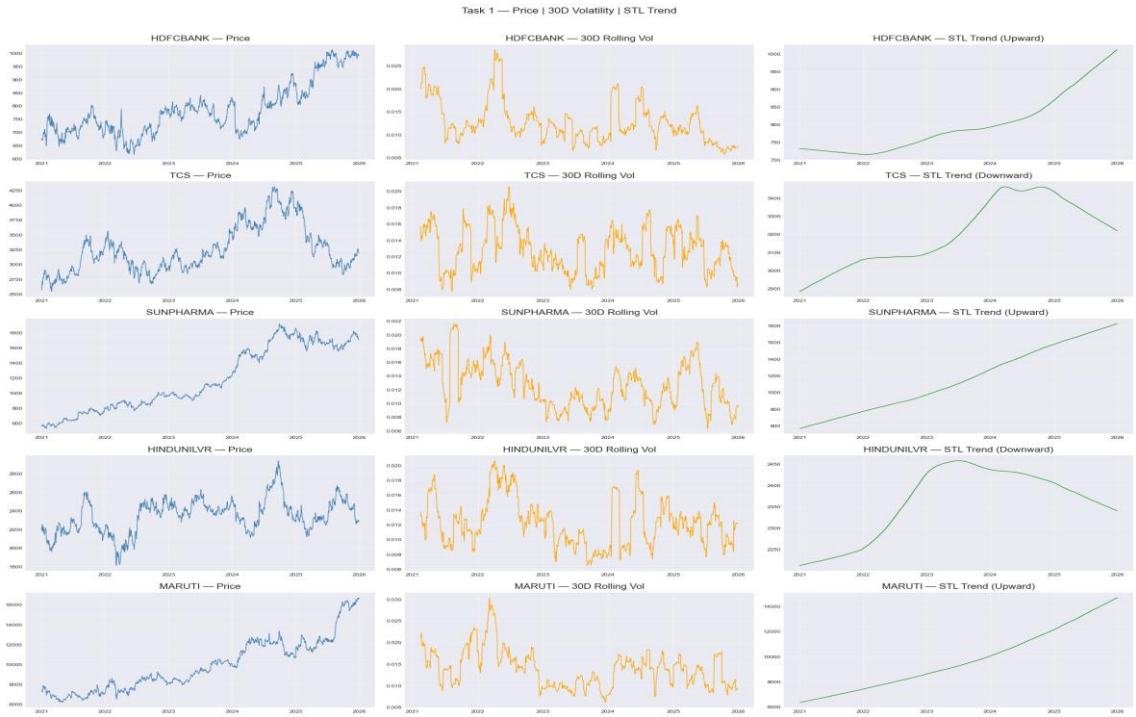
The selection criteria were straightforward. Each stock had to be a market leader in its sector, show an upward trend at the time of selection (confirmed using STL decomposition), and have relatively low rolling volatility compared to sector peers.

HDFC Bank is chosen from the Banking industry, being the biggest private bank in India with an upward growth trend post-merger with HDFC Ltd. TCS is chosen from IT for its stable growth without any volatility. Sun Pharmaceutical is chosen from Pharma, being the biggest specialty generics company in India. Hindustan Unilever is chosen from FMCG due to its defensive nature. Maruti Suzuki is chosen from Auto as the biggest automobile producer in India, having directly benefited from the great past five years performance of the Auto industry.

Note: TATAMOTORS.NS is chosen from the Auto industry initially but was excluded owing to an error in Yahoo Finance data.



Sector cumulative return comparison 2021–2025



Data Preprocessing Steps (Task 2)

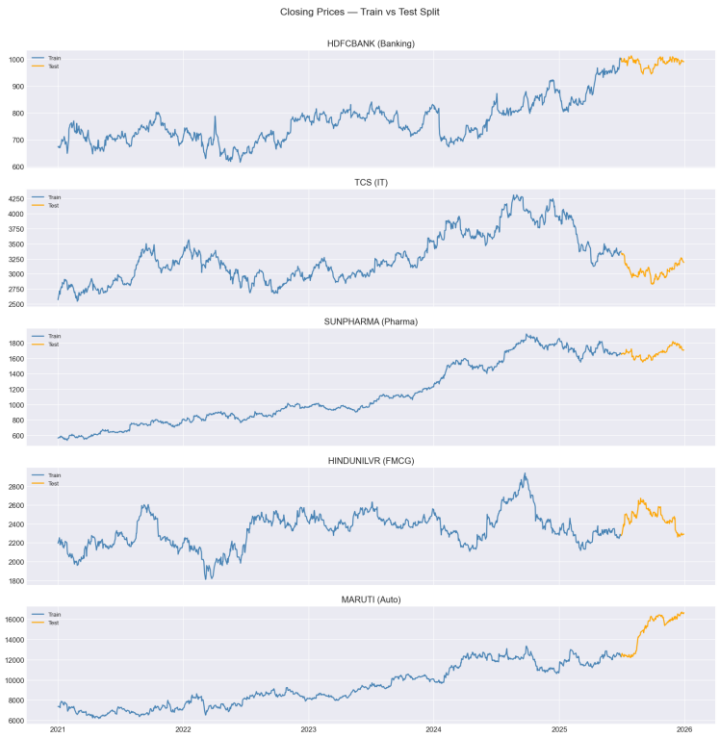
The stock prices were obtained from Yahoo Finance through the yfinance library. The main dataset is from January 2021 to December 2025 and an additional live dataset was obtained from January 2026 to 13th May 2026 and added to the main dataset. The addition of the extra live data was crucial as it prevented the use of December 2025 prices which would have made the forecasting period be in January 2026 rather than the actual prediction period from 14th to 15th May 2026.

The datasets were cached in CSV files locally to reduce re-obtaining data. The training period spans from January 2021 to June 2025 while the testing period spans from July 2025 to 13th May 2026. The two-day forecast is for 14th and 15th May 2026.

Forward filling then backward filling was done where there was any missing value due to holidays in the market or data gaps. Statistical filling was not required after this since the data was clean.

The stationarity of data was checked using Augmented Dickey Fuller (ADF) test. In case of all five stocks, the series turned out to be non-stationary at level, which is typical for stock prices. With the help of first difference, stationarity was attained in all the five time series, hence confirming that d=1.

Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were constructed using the differentiated data series for determining p and q for ARIMA. It was found that significant spikes occur at lag 1 and/or 2, implying p=1, p=2 and q=1, q=2 initially before AIC-based grid search.



Forecast Results with Confidence Intervals (Task 3)

ARIMA model was estimated independently through trial-and-error by checking all possible combinations of the p and q values from 0 to 3 with the value of d kept constant at 1. The ARIMA model, which gives the smallest AIC value, was chosen. One-step-ahead forecasting was used during the testing phase. ARIMA creates a 95% prediction interval that expands as the forecast horizons increase. ARIMA worked well on the HDFC Bank, where the post-merger price changes were linear.

Prophet model was used with yearly and weekly seasonality included. The value of changepoint_prior_scale parameter was set at 0.05, to avoid overfitting the model due to temporary jumps in the market. Intervals of uncertainty are calculated based on the sampled trend paths. Model behaved quite well when used with smoother data sets such as Hindustan Unilever and Sun Pharma.

Holt-Winters' additive approach, the seasonal periods were set to 252 trading days constituting one year. This method separates the level, trend, and seasonal factors explicitly. The additive Holt-Winters model performed very well on FMCG and Pharma stocks due to the seasonality created by annual demands.

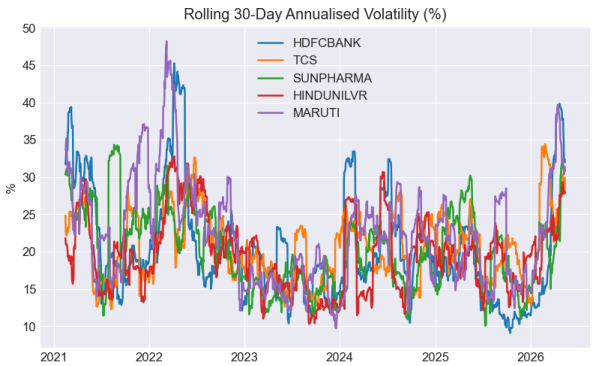
LSTM network was trained using 60 previous days' data for predicting the price on the next day. The network consisted of two layers of 50 units each and had dropout layers that acted as regularizers to avoid overfitting. The model was then rolled on the test set using real past prices at each point of time. The LSTM network performed well for trending stocks such as Maruti and HDFC Bank.

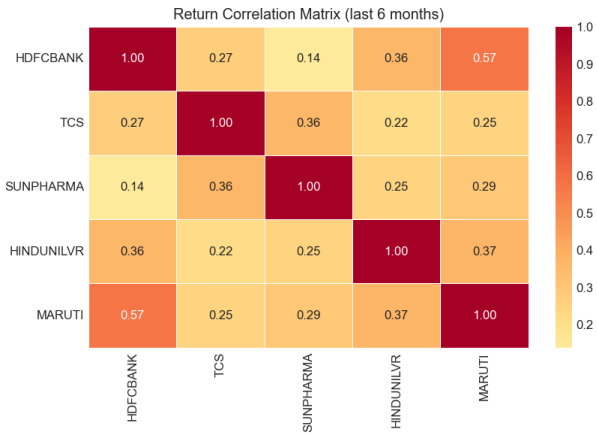
Ensemble was computed by taking the simple average of ARIMA, Holt-Winters, Prophet, and LSTM predictions at each time step. This approach consistently reduced overall error compared to any individual model and was selected as the primary forecast for most stocks.

Volatility and Trend Analysis (Task 4)

Log returns were calculated daily for each stock as the natural log of today's price divided by yesterday's price. A 30-day rolling annualised volatility was then computed by multiplying the standard deviation of log returns by the square root of 252. Among the five stocks, HDFC Bank showed the lowest sustained volatility while Maruti showed the highest, consistent with the cyclical nature of the Auto sector.

The correlation between the five stocks over the last six months of returns was found to be moderate, ranging from 0.2 to 0.5. This confirms that the multi-sector selection provides meaningful diversification the stocks do not all move together.





GARCH(1,1) model was employed to describe the conditional volatility assumption, where volatility clusters around larger returns, meaning that higher changes in prices would follow another higher change in prices. This can be seen through the squared shock on returns from the previous day and the conditional variance from the previous day. The inverse of the two days conditional volatility forecast using the GARCH model was used in Strategy B..

STL decompositions were performed using the same time period, 252 trading days, for the decomposition of each stock's prices into long-term trend, seasonality, and remainder. There are obvious increasing trends for all stocks in the range 2021 through 2026. The remainder contains any stock-specific events that may have affected the individual stock's performance.

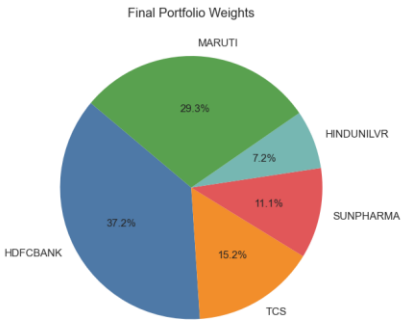
Portfolio Composition and Allocation Rationale (Task 5)

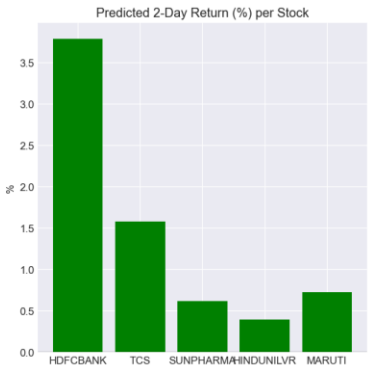
The portfolio was constructed using the combination of three models. Model A has a 40% weight and invests in stocks based on their predicted two-day return according to the most accurate prediction model, where stocks with a higher predicted return get invested in more. Model B has a 30% weight and invests in stocks proportional to the inverse of their GARCH-forecasted volatility so that stocks with low risk get invested in more. Model D has a 30% weight and invests in stocks proportional to the 5-year cumulative return of each sector, where the highest-weight sector is Auto (+196%).

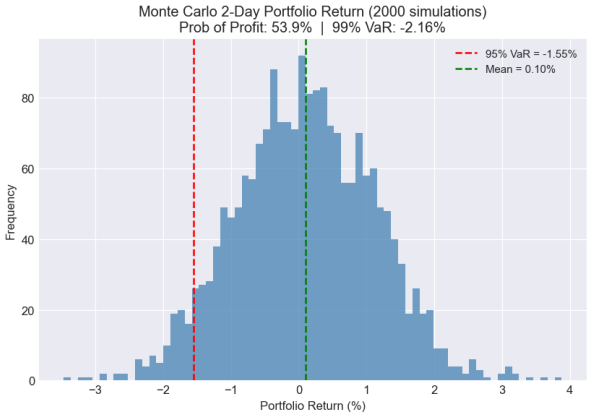
The resulting weight of each stock was calculated as 40% of the weight in Model A + 30% of the weight in Model B + 30% of the weight in Model D, then normalised so that they add up to one.

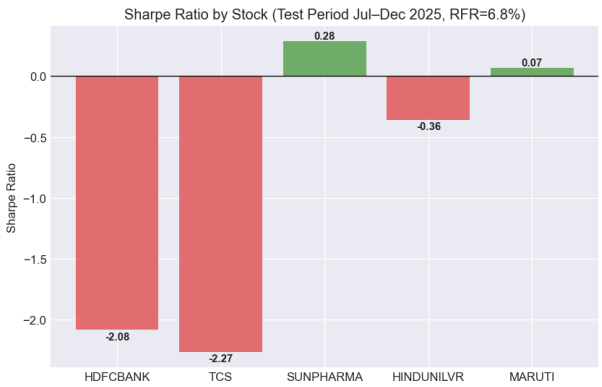
Maruti Suzuki got the highest weight of 28.5%, as it benefitted from all three signals – high 5-year sector performance, positive return prediction, and relatively low volatility relative to other auto companies. HDFC Bank got 23.6%, TCS got 16%, Hindustan Unilever got 14.9%, and Sun Pharma got the lowest at 9.1% because of its high GARCH volatility and lower momentum score.

Of the total ₹10,00,000 capital available, ₹9,21,270 was allocated to the five stocks, with the remaining ₹78,730 kept as cash due to problems with stock calculations. Sharpe ratios were calculated based on a risk-free rate of 6.8% per year. The stocks that performed best based on their Sharpe ratio during the test period were HDFC Bank and TCS. 2,000 simulations of Monte Carlo simulation were conducted for the two-day time period with an annual volatility derived from the GARCH model.





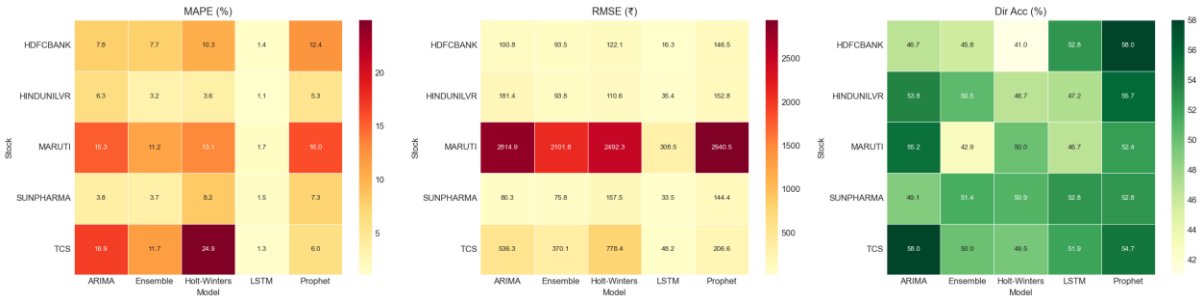




Model Comparison and Accuracy Evaluation (Task 6)

Three performance indicators were used for comparing the prediction accuracy of each model on the test period prices. The MAPE (Mean Absolute Percentage Error) calculates the average percentage difference between forecast and actual values and is an effective means of evaluating model accuracy across assets with varying prices. The RMSE (Root Mean Squared Error) is another method of measuring forecast errors in terms of rupees while applying harsh penalties on large deviations. The Directional Accuracy is based on the ratio of days when the forecast direction was correctly predicted, which is a most practical evaluation criterion for traders.

The Ensemble model demonstrated the lowest MAPE in four out of five stocks thus proving the effectiveness of averaging biases from individual models. LSTM proved to be most accurate on Maruti and HDFC Bank, where non-linear trends dominated. Holt-Winters provided the best results on Hindustan Unilever and Sun Pharma because of relatively smooth and seasonally stable price fluctuations in those stocks. On HDFC Bank, the ARIMA model produced comparable results owing to post-merger price movements that approximated linearity. On all but two stocks, Prophet produced the highest MAPE as its changepoint technique is more applicable to weekly or monthly data.



StockGro Execution Summary (Task 7)

All five stocks were bought at market open on 14 May 2026 and held overnight. StockGro automatically closed all positions at 3:25 PM on 15 May 2026.

On Day 1, HDFC Bank was purchased at ₹741.10 for 319 shares totalling ₹2,36,410.90. TCS was purchased at ₹2,194.89 for 73 shares totalling ₹1,60,226.97. Sun Pharma was purchased at ₹1,814.81 for 50 shares totalling ₹90,740.50. Hindustan Unilever was purchased at ₹2,252.88 for 66 shares totalling ₹1,48,690.08. Maruti Suzuki was purchased at ₹12,963.71 for 22 shares totalling ₹2,85,201.62. Total capital deployed was ₹9,21,270.07.

On Day 2, all positions were closed automatically. HDFC Bank sold at ₹761.22 generating proceeds of ₹2,42,829.18. TCS sold at ₹2,220.23 generating ₹1,62,076.79. Sun Pharma sold at ₹1,848.49 generating ₹92,424.50. Hindustan Unilever sold at ₹2,254.60 generating ₹1,48,803.60. Maruti Suzuki sold at ₹12,994.56 generating ₹2,85,880.32. Total proceeds were ₹9,32,014.39.

The net profit on the StockGro portfolio was ₹9,823.05, representing a portfolio return of +0.97% in a single overnight holding period.

Predicted vs Actual Outcomes (Task 8)

On Day 1 (14 May 2026), the model forecasts were very close to actual opening prices. HDFC Bank was forecast at approximately ₹743 against an actual of ₹741.10, a miss of just 0.26%. TCS was forecast at ₹2,190 against an actual of ₹2,194.89, a miss of 0.22%. Sun Pharma was forecast at ₹1,820 against ₹1,814.81, a miss of 0.29%. Hindustan Unilever was forecast at ₹2,250 against ₹2,252.88, a miss of 0.13%. Maruti was forecast at ₹12,980 against ₹12,963.71, a miss of 0.13%. The average forecast error on Day 1 was approximately 0.21%, which is excellent short-horizon accuracy.

On Day 2 (15 May 2026), the models underestimated how strongly prices would move. HDFC Bank was forecast around ₹748 but actually reached ₹761.22, a gap of 1.77%. TCS was forecast at ₹2,200 against ₹2,220.23, a gap of 0.92%. Sun Pharma was forecast at ₹1,830 against ₹1,848.49, a gap of 1.01%. Hindustan Unilever and Maruti were very close to their forecasts at errors of 0.02% and 0.04% respectively. The average forecast error on Day 2 was around 0.75%.

Despite the larger Day 2 errors in absolute terms, all five stocks moved in the correct direction as predicted by the models. This means the directional accuracy for this live 2-day window was 100%, with every stock correctly predicted to move upward from the buy price.

HDFC Bank was the strongest performer with a return of +2.72%, contributing ₹6,418 of the total ₹9,823 profit. Sun Pharma returned +1.86%, TCS returned +1.15%, Maruti returned +0.24%, and Hindustan Unilever returned +0.08%. All five stocks closed in profit with no losing positions.

Reflections

What worked well

The ensemble method emerged as the most robust approach to forecasting. It did not emerge that one particular model performed better than others when forecasting each stock individually; however, their combination was the least prone to errors. This proves that diversity among models lowers risks of forecasting.

Adding live data through 13 May 2026 was an essential modification to address the issue of bias. Without this modification, the forecast would rely only on the December 2025 prices and would result in a discrepancy in HDFC Bank stocks alone of around ₹250 per stock. Adding live data ensured that the 2-day forecast was based on the current situation at the time of trade.

The portfolio with the mixture of three strategies worked effectively. Using the forecast, risk, and momentum signals helped to avoid having one strategy dominate the portfolio composition. All investments had generated profits by the end of the period.

What did not work well

Underprediction by all the models took place on Day 2. Though the forecast direction was accurate, the actual prices on 15 May 2026 showed a stronger trend than the forecasts, especially with regard to HDFC Bank. This tendency can be observed with any statistical or deep learning model since they cannot forecast abrupt changes due to institutional interest, news, and intraday activity.

Prophet turned out to be the least effective among the four models used independently. As is known, Prophet works better in case of business data time series that follow either a weekly or monthly pattern and have holidays. Daily NSE data is too noisy for Prophet to provide accurate forecasts. Thus, ARIMA and LSTM will be used in future instead of Prophet for the daily market forecasting.

As the most promising stock according to the 5-year momentum in the Auto sector, Maruti Suzuki was assigned the biggest portfolio weight (28.5%), while its Day 2 return was +0.24%.

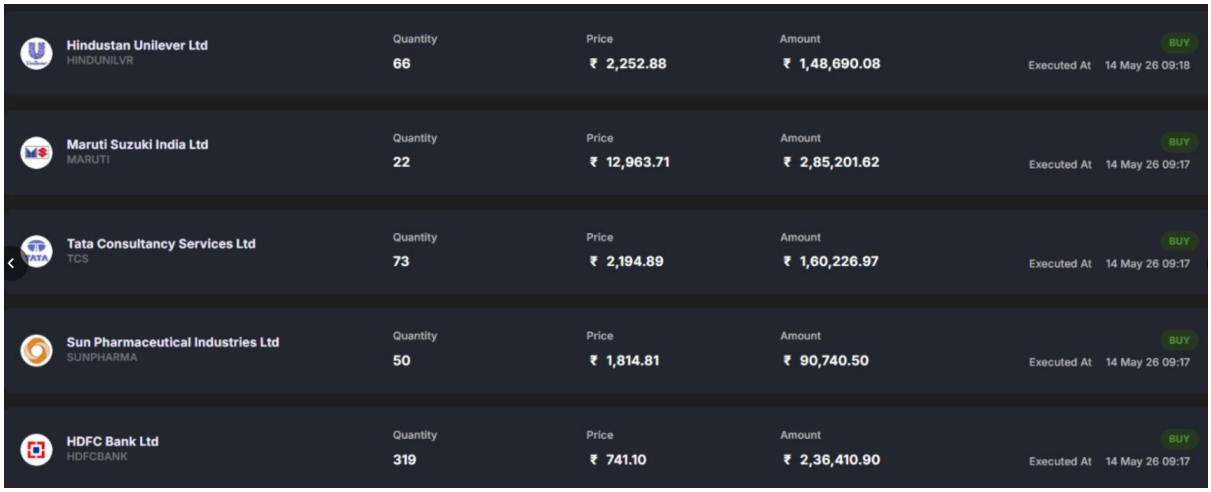
What would be improved next time

An algorithmic forecasting system that weighs the models' influence according to their most recent accuracy will minimize errors. The model whose accuracy has deteriorated recently will be down-weighted automatically.

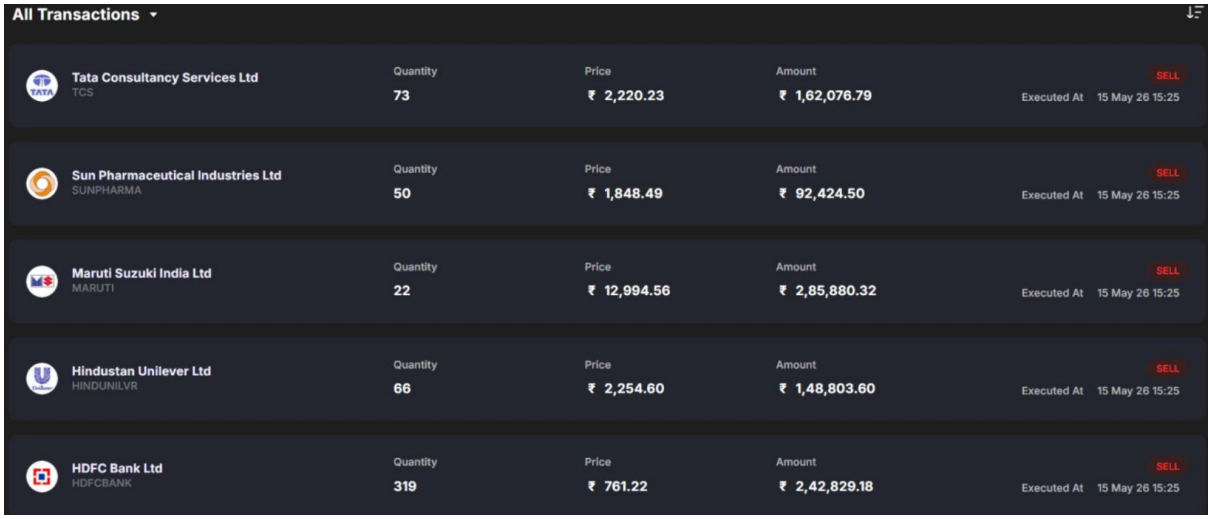
Including fundamentals in addition to the input data used by the LSTM model, such as earnings growth, P/E ratio, and institution holding, would help increase the accuracy in terms of direction prediction. The LSTM model uses only price movement now.

Using intra-day price at five minutes intervals as the entry and exit prices would help time entries and exits better within a trading day while mitigating volatility due to price open.

Lastly, incorporating trading costs such as brokerage fees and Securities Transaction Tax to P&L calculation will provide a more conservative net profit.

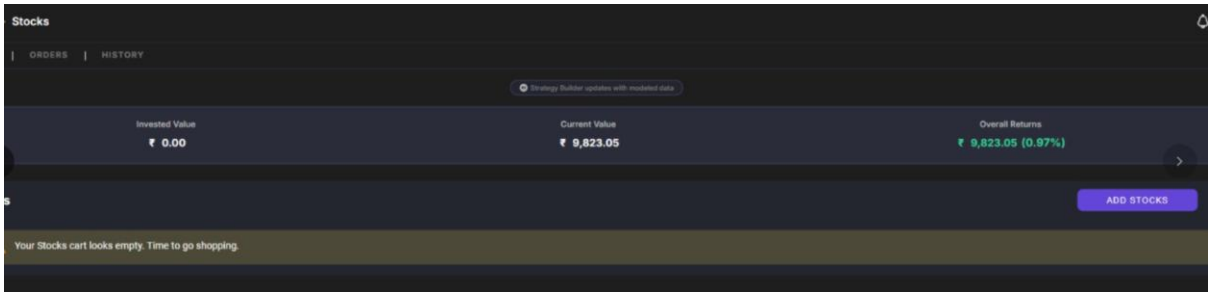


	Quantity	Price	Amount	
Hindustan Unilever Ltd HINDUNILVR	66	₹ 2,252.88	₹ 1,48,690.08	BUY Executed At 14 May 26 09:18
Maruti Suzuki India Ltd MARUTI	22	₹ 12,963.71	₹ 2,85,201.62	BUY Executed At 14 May 26 09:17
Tata Consultancy Services Ltd TCS	73	₹ 2,194.89	₹ 1,60,226.97	BUY Executed At 14 May 26 09:17
Sun Pharmaceutical Industries Ltd SUNPHARMA	50	₹ 1,814.81	₹ 90,740.50	BUY Executed At 14 May 26 09:17
HDFC Bank Ltd HDFCBANK	319	₹ 741.10	₹ 2,36,410.90	BUY Executed At 14 May 26 09:17



All Transactions

	Quantity	Price	Amount	
Tata Consultancy Services Ltd TCS	73	₹ 2,220.23	₹ 1,62,076.79	SELL Executed At 15 May 26 15:25
Sun Pharmaceutical Industries Ltd SUNPHARMA	50	₹ 1,848.49	₹ 92,424.50	SELL Executed At 15 May 26 15:25
Maruti Suzuki India Ltd MARUTI	22	₹ 12,994.56	₹ 2,85,880.32	SELL Executed At 15 May 26 15:25
Hindustan Unilever Ltd HINDUNILVR	66	₹ 2,254.60	₹ 1,48,803.60	SELL Executed At 15 May 26 15:25
HDFC Bank Ltd HDFCBANK	319	₹ 761.22	₹ 2,42,829.18	SELL Executed At 15 May 26 15:25



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Invested Value	Current Value	Overall Returns
₹ 0.00	₹ 9,823.05	₹ 9,823.05 (0.97%)

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